

Iron catalysts supported on carbon nanotubes for Fischer–Tropsch synthesis: Effect of catalytic site position

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ABSTRACT

In order to study the effects of catalytic site position on Fischer–Tropsch (FT) reactions, a method was developed to control the position of the catalytic sites on either inner or outer surface of carbon nanotubes (CNTs). TEM analyses revealed that more than 70–80% of iron oxide particles can be controlled to be positioned at inner or outer surface of the nanotubes. Based on H₂-TPR analysis, deposition of iron oxide inside the nanotube pores resulted in easier reduction of the oxide at a lower temperature (from 418 to 381 °C). Catalytic performances of the catalysts in terms of FT experiment were tested in a fixed-bed reactor. According to the results of FT experiments, both catalysts showed similar initial %CO conversion (~90%). However, the catalyst with catalytic sites inside the pores exhibited higher selectivity to heavier hydrocarbons. In addition, deposition of catalytic sites on interior surface of the nanotubes resulted in a more stable catalyst, while its counterpart experienced deactivation within a period of 125 h due to catalytic sites sintering. It is concluded that encapsulation of catalytic sites inside the nanotubes prevents the catalytic site agglomeration.

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1. Introduction

Fischer–Tropsch synthesis (FTS) is a potentially promising option for environmentally friendly production of chemicals and fuels from biomass, coal and natural gas [1,2]. In industrial applications, a high-performance catalyst plays an essential role. In FTS process, the catalyst activity and selectivity are influenced by nature and structure of support, nature of metal, metal dispersion, metal loading, and catalyst preparation method [3,4]. Most studies on FTS catalysts have been carried out with the metals supported on silica, alumina or titania. In recent years, other families of supports with carbonaceous base in the form of activated carbon and carbon nanotubes (CNTs) have been investigated for FT reactions [6–20].

Carbon nanotubes with unique properties such as uniform pore size distribution, meso and macro pore structure, inert surface properties, and resistance to acid and base environment can play an important role in a large number of catalytic reactions [18].

Abbreviations: BET, Brunauer, Emmett and Teller; CNT, Carbon nanotubes; FTS, Fischer–Tropsch synthesis; ICP, Inductively coupled plasma; SEM, Scanning electron microscopy; SL, Standard liter; TEM, Transition electron microscopy; TOS, Time on-stream; TPR, Temperature programmed reduction; WGS, Water-gas shift; XRD, X-ray diffraction; In-Fe/CNT, Iron catalysts deposited on interior surface of carbon nanotubes; Out-Fe/CNT, Iron catalysts deposited on exterior surface of carbon nanotubes.

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Carbon nanotubes (CNTs) are different compared to other supports in that they have graphene layers with a tubular morphology. Theoretical studies reveal that deviation of the graphene layers from planarity causes π -electron density to shift from the concave inner surface to the convex outer surface, leading to an electron-deficient interior surface and an electron-enriched exterior surface [19–21]. This can influence the structure and electronic properties of substances in contact with either surface. In addition, studies on non-catalytic gas-phase reactions suggest that confinement within small channels could increase the density of reactants, and hence create a locally higher pressure [22] which will favor syngas conversion. However, a possible drawback of deposition of catalytic sites inside the narrow channels is higher mass transfer restriction.

Chen et al. [19] studied the confinement effect of iron particles inside the nanotubes channels for FT reactions. They used closed cap CNTs for deposition of iron particles on exterior surface of nanotubes using impregnation method. In order to prepare iron catalyst inside the nanotubes, they used strong acid pre-treatment to open the CNTs' cap. Therefore, their catalysts experienced different pre-treatment procedure which may affect the catalyst performance [5]. According to their report, the catalytic performance of the iron catalysts supported either inside or outside of the nanotubes' channel is similar at pressure less than 2 MPa. However, at higher pressure (5 MPa), the yield of C₅+ hydrocarbons over the encapsulated iron catalyst is twice that over iron catalyst outside CNTs.

Nomenclature

P	Pressure (MPa)
T	Temperature ($^{\circ}\text{C}$)
%CO	Percent of CO conversion (mol%)

In this research, we have introduced a method to manipulate the catalytic site position on interior or exterior surface of the CNTs through same pre-treatment procedure. The main objective of the current work is to study the effect of position of the active metal on FT reaction.

2. Experimental

A batch of carbon nanotubes was purchased from Mknano-MWCNTTM. The CNT sample was treated in nitric acid (60 wt%) at 110 $^{\circ}\text{C}$ for 16 h. The catalysts were prepared according to incipient wetness impregnation method. For the preparation of the catalyst with iron particles inside the nanotube pores (in-Fe/CNT) with 12 wt% Fe, a specified amount of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was dissolved in deionised water. The amount of solution was equal to total pore volume of the CNT sample ($0.62 \text{ cm}^3/\text{g}$). In this case, the metal solution fills the inner pores of the CNTs due to capillary effects. In order to prepare the catalysts with catalytic sites on the exterior surface (out-Fe/CNT), the acid treated CNT sample was wetted with pure water (the amount of water was equal to the pore volume of the CNT sample). As a result, the inner pores of the CNTs were initially filled with water. Afterward, the metal salt solution was added to the support. In this case, there was no capillary force for the metal solution to move into the nanotubes' pores. However, there might be a slight diffusion of metal salt solution into the water-filled-pores [23]. Subsequently, the impregnated support sample was immediately moved into the drying oven. After drying at 120 $^{\circ}\text{C}$ and calcination at 400 $^{\circ}\text{C}$ for 3 h, the catalysts were characterized by nitrogen adsorption, ICP, TPR, XRD, SEM and TEM.

The surface area, pore volume, and average pore diameter of the CNTs and catalysts were measured by an ASAP-2000 Micromeritics system. The samples were degassed at 200 $^{\circ}\text{C}$ for 2 h under 50 m Torr vacuum before measurements.

XRD analysis was performed using a Philips PW1840 X-ray diffractometer with monochromatized $\text{Cu}/\text{K}\alpha$ radiation. The crystallite diameter was determined by substituting the half-width of a chosen peak into the Debye–Scherrer equation.

The reduction behavior of the catalyst precursors was studied by temperature programmed reduction (TPR) using a CHEMBET-3000 equipped with a thermal conductivity detector. 0.1 g of catalyst was placed in U-shaped quartz tube. A 5% hydrogen/nitrogen mixture was introduced (flow rate = 36 cm^3 (STP)/min)

and the furnace temperature was ramped from room temperature to 1173 K at 10 K/min.

Morphology of the samples was studied by transmission electron microscopy (TEM). Sample specimens for TEM studies were prepared by ultrasonic dispersion of the catalysts in ethanol, and the suspensions were dropped onto a copper grid. TEM micrographies were carried out using a Hitachi H-7500 (120 kV). Several TEM micrographs were recorded for each sample and were analysed to determine the particle size distribution. A Phillips SEM-505 scanning electron microscope operating at 300 kV in SE display mode was also used for SEM micrographies. For characterization prior to analysis, all the samples were gold coated in a sputter coating unit (Edwards Vacuum Components Ltd., Sussex, England).

The Fischer–Tropsch synthesis was performed in a fixed-bed micro reactor. Prior to CO hydrogenation, in situ reduction was conducted according to the following procedure. The catalyst (1 g) was placed in the reactor and diluted with 7 g silicon carbide. Then pure hydrogen was introduced at a flow rate of 30 ml/min. The reactor temperature was increased from room temperature to 380 $^{\circ}\text{C}$ at a rate of 1 $^{\circ}\text{C}/\text{min}$ and maintained at this activation condition for 14 h. After the activation period, the reactor temperature was reduced to 270 $^{\circ}\text{C}$ under flowing hydrogen. Hydrogen and syngas flow rates were controlled by Brooks 5850 mass flow controllers. Argon was used as internal standard gas in the reactor feed. The mixed gases (30%CO, 60% H_2 , 10%Ar) entered through the top of the fixed-bed reactor. The temperature of the reactor was controlled using a PID temperature controller. Synthesis gas with a space velocity of $2000 \text{ cm}^3(\text{STP})/(\text{h g})$ was introduced and the reactor pressure was increased to 2 MPa. Products were continuously removed from the vapor and passed through two traps. The pressure of the uncondensed gaseous product stream was reduced to atmospheric pressure. The composition of the outlet gas stream was measured using an on-line GC-2014 Shimadzu gas chromatograph. The contents of the traps were removed every 24 h, the hydrocarbon and water fractions were separated, and analysed by Varian 3400 GC. Catalytic activity, product selectivity and stability of the catalyst were monitored during reaction period of 125 h.

3. Results and discussions

3.1. Catalysts characterizations

SEM images of the crude CNTs revealed that the sample contained solely CNTs and no carbonaceous impurities were observed (Fig. 1a). The fresh CNT sample contained woven nanotubes with high aspect ratios. TEM micrographs of the fresh and acid treated CNT supports are shown in Fig. 1b and c, respectively. Fig. 1b shows that the fresh CNTs have uniform

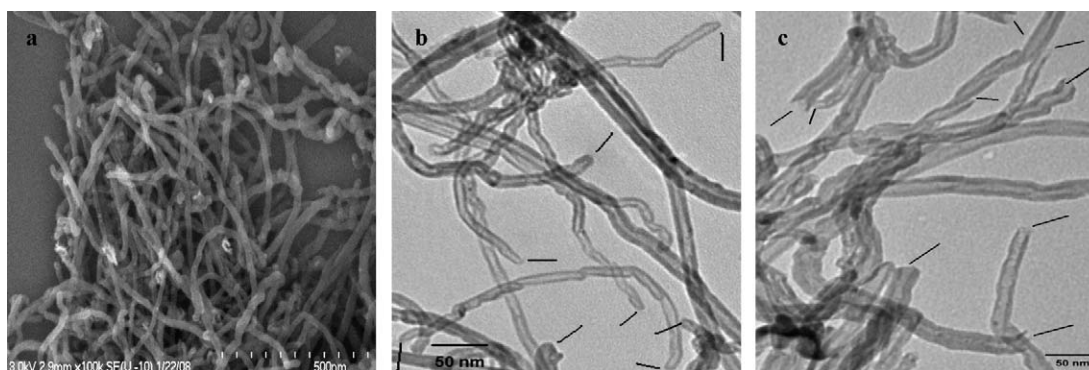


Fig. 1. (a) SEM micrograph of fresh CNT support, (b) TEM image of fresh CNT support, (c) TEM image of acid treated CNT support (lines show close and open capped nanotubes).

Table 1

Metal content and surface properties of the fresh, pre-treated supports and iron catalysts.

Catalyst	Metal content (%) ^a	BET surface area (m ² /g)	Total pore volume (cm ³ /g)
Fresh CNT	–	181	0.55
Acid treated CNTs	–	234	0.62
In-Fe/CNT	11.6	199	0.51
Out-Fe/CNT	11.9	211	0.59

^a Determined by ICP analysis.

nanotubes and their inner and outer diameters vary between 5–12 nm and 10–25 nm, respectively. In the case of fresh CNTs, most of the nanotubes' caps are closed while acid treatment resulted in opening nanotubes' caps as shown in Fig. 1c. This is also verified with nitrogen adsorption analysis. The BET surface area increased from 181 to 234 m²/g (Table 1) with the acid treatment. The pore volume of the CNTs was also increased from 0.48 to 0.62 cm³/g after the acid treatment.

The position of the iron particles inside or outside of the CNTs was verified using several TEM images from each of the catalyst samples. Fig. 2 shows sample TEM images of in-Fe/CNT and out-Fe/CNT catalysts. Dark spots represent the iron oxides, which are attached inside or outside of the nanotubes. In the case of in-Fe/CNT, vast majority of the iron particles (80%) are distributed in the inner pores of CNTs (Fig. 2a). This can be attributed to carbon nanotubes' tubular morphology which can induce capillary forces during the impregnation process. For out-Fe/CNT catalyst, most of the iron particles (70%) were attached to the exterior surface of CNTs. This is because, their inner pores are initially filled with water and iron salt solution can only contact the outside of the nanotubes and thus, the majority of iron oxide particles are attached to the exterior surface of the CNTs (Fig. 2a and b).

Table 1 shows the metal content for the catalysts. ICP analyses of the catalysts revealed that the metal contents of the catalysts were fairly similar and close to the target metal content of 12 wt% Fe. Table 1 also shows the results of surface area measurements of in-Fe/CNT and out-Fe/CNT. For in-Fe/CNT and out-Fe/CNT catalysts, the loading of 12% Fe decreased the surface area to 199 and 211 m²/g, respectively. The results show that the BET surface area of the catalysts is lower than that of the supports indicating pore blockage due to iron loading on the supports. However, comparison between BET surface area of in-Fe/CNT and out-Fe/CNT catalysts reveals that the extent of pore blockage in the in-Fe/CNT is higher than that in out-Fe/CNT.

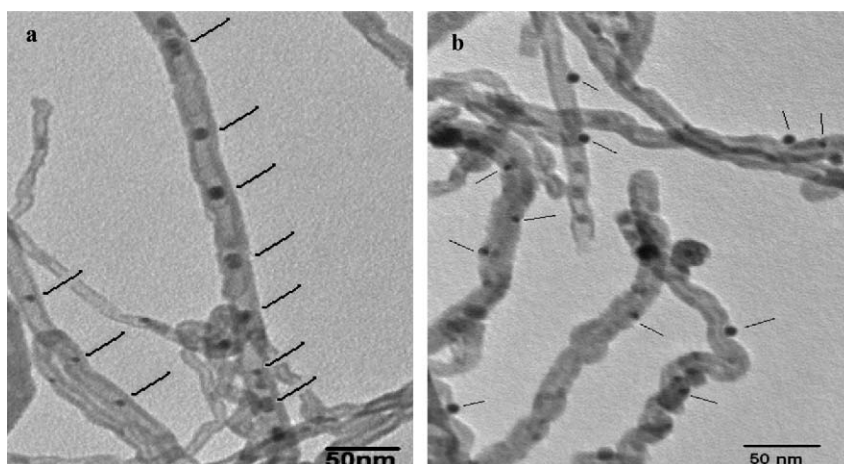


Fig. 2. TEM image of (a) in-Fe/CNT, (b) out-Fe/CNT (lines show iron oxide particles).

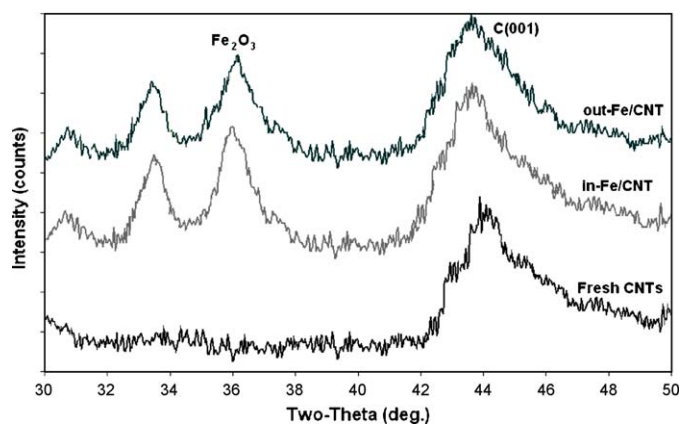


Fig. 3. XRD patterns of in-Fe/CNT, out-Fe/CNT and fresh CNTs.

Fig. 3 shows XRD patterns of the CNT support as well as calcined catalysts. Both of the catalysts show similar XRD patterns between 2 θ values of 10° and 50°. The diffraction peaks match very well with the standard Hematite (Fe₂O₃) phase. For fresh CNT support, in-Fe/CNT and out-Fe/CNT catalysts, the peaks at 2 θ of 26 and 43° correspond to graphite layers (multiwall carbon nanotubes), while the other peaks in the spectrum of catalysts are related to different crystal planes of Fe₂O₃. The peak at 35.7° is the most intense peak of Fe₂O₃ in the XRD spectrum of the catalysts.

Table 2 shows the average Fe₂O₃ particle size on the catalysts calculated from XRD spectrum and Debye–Scherrer equation at the most intense peak of 35.7°. The data reveal that both catalysts possess similar sizes of iron oxide particles (7–9 nm). To compare the particle size calculated based on XRD with TEM analysis, the size distribution of the particles is shown in Table 2. There is a good agreement between the data for average particle size calculated based on XRD and TEM size distribution.

Table 2

Particle sizes (iron oxide) based on XRD and TEM analysis.

Catalyst	$d_{\text{Fe}_2\text{O}_3}^a$ (nm)	$d_{\text{Fe}_2\text{O}_3}^b$ (nm)
In-Fe/CNT	9 ± 1	6–11
Out-Fe/CNT	7 ± 1	5–9
Used in-Fe/CNT	11 ± 2	6–12
Used out-Fe/CNT	16 ± 2	6–24

^a Based of XRD analysis.

^b Based of TEM analysis.

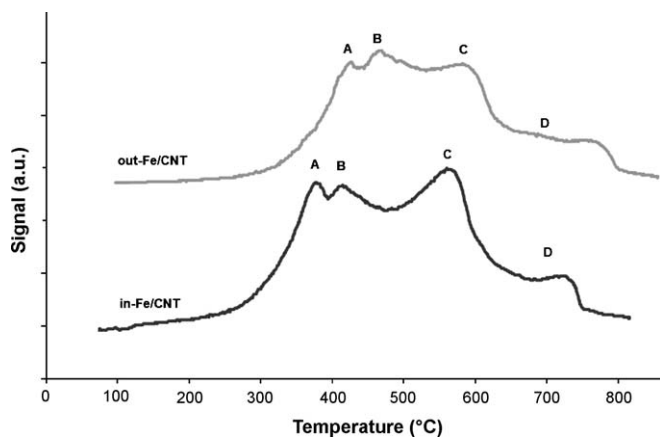
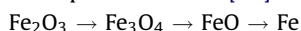


Fig. 4. H₂-TPR profiles for in-Fe/CNT and out-Fe/CNT catalysts.

TPR analyses were performed to evaluate the reducibility of the studied catalysts. TPR patterns of in-Fe/CNT and out-Fe/CNT catalysts are shown in Fig. 4. Four peaks (A, B, C and D) can be observed on the TPR profile. Generally, the reduction of iron oxides takes place as below [11]:



The first peak, A, can be assigned to the reduction of Fe₂O₃ to Fe₃O₄. The second peak, B, can be assigned to the subsequent reduction of Fe₃O₄ to FeO. Peak C, observed at 500–600 °C, can be related to the reduction of FeO to metallic Fe. Peak D, corresponds to gasification of CNTs at a temperature higher than 600 °C. Table 3 shows reduction temperature for both catalysts. According to the reduction temperature, deposition of iron oxide particles inside the nanotubes results in a decrease in the temperature of the first TPR peak from 418 to 381 °C and that of the second and third from TPR peak from 470 to 411 °C and 591 to 567 °C, respectively. As reported by other researchers [19], the confinement of iron oxide inside the CNT pore resulted in easier reduction at lower temperature. It can be concluded that iron oxide interacts with the interior CNT wall differently from that with the exterior wall. It

Table 3
Reduction temperature (°C) based on H₂-TPR analysis.

Catalyst	Peak A	Peak B	Peak C
In-Fe/CNT	381	411	567
Out-Fe/CNT	418	470	591

has been postulated that the electron deficiency of the interior CNT surface is possibly responsible for this phenomenon.

3.2. Used catalysts characterization

XRD and TEM analyses were conducted to study the particle size of the used catalysts after 125 h of FTS reaction at 270 °C. The diffraction peaks match very well with magnetite (Fe₃O₄) phase (The XRD spectra are not shown here). The corresponding peaks for iron carbide phase (Fe₅C₂) appeared in the XRD spectra of the used in-Fe/CNT catalyst while there were no noticeable peaks in the used out-Fe/CNT for iron carbide phase. This evidence showed that for the in-Fe/CNT catalyst, the iron carbide content was higher and more stable during the FT reactions.

Based on XRD peak broadening, the particle size calculation revealed particle growth for the used out-Fe/CNT catalysts. In the case of out-Fe/CNT, a sharp peak appeared at 2θ angle of 35.6° indicating particle growth, whereas, for in-Fe/CNT catalyst, the particle growth is not significant. The out-Fe/CNT catalyst experienced site agglomeration increasing in size from 7 to 16 nm whereas in the case of in-Fe/CNT, the particle sintering changed the particle size from 9 to 11 nm (see Table 2). Fig. 5 shows TEM images of the used in-Fe/CNT and in-Fe/CNT. As can be seen in Fig. 5a, the iron oxide particles inside the channels do not experience particle agglomeration. This phenomenon can be related to the interaction of the catalytic sites with the inner surface of the pores and perhaps to the spatial restriction of the CNT channels. However, most of the iron species on the exterior surface agglomerated (Fig. 5b and c) resulting in lower catalytic dispersion under FT reactions. From XRD data and particle size distribution, it can be concluded that the deposition of the catalytic sites inside the nanotube pores results in relatively more stable catalyst than that deposited on exterior surface of the nanotubes.

In order to study the change in particle size distributions and mechanism for the catalytic site agglomeration, several TEM images from fresh and used in-Fe/CNT and out-Fe/CNT catalysts were taken and analysed. The results are shown in Fig. 6a and b. As can be seen in Fig. 6a, for in-Fe/CNT catalyst, there was a slight shift toward larger particle size with the maximum of 11 nm. However, in the case of used out-Fe/CNT catalyst, the particle size distribution shifted considerably toward larger particles with an average of 18 nm and maximum particle size of 24 nm.

In order to describe the catalytic site agglomeration, two mechanisms namely, atom migration (Ostwald ripening) and crystallite migration (coalescence) are generally considered. According to the Ostwald ripening theory, metal atoms are released from one metal particle and attached to another metal particle. The difference in surface energy is the driving force for this

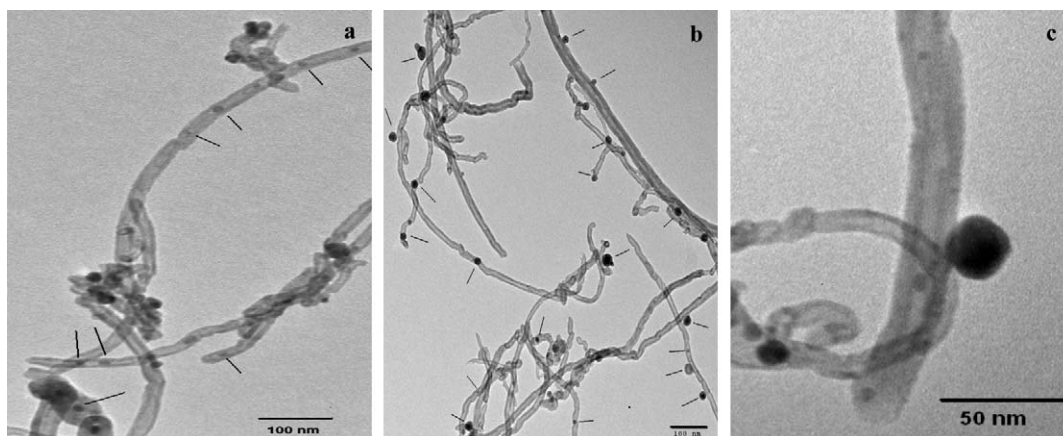


Fig. 5. TEM image of (a) used in-Fe/CNT, (b) used out-Fe/CNT (lines point to iron oxide particles) (c) used out-Fe/CNT.

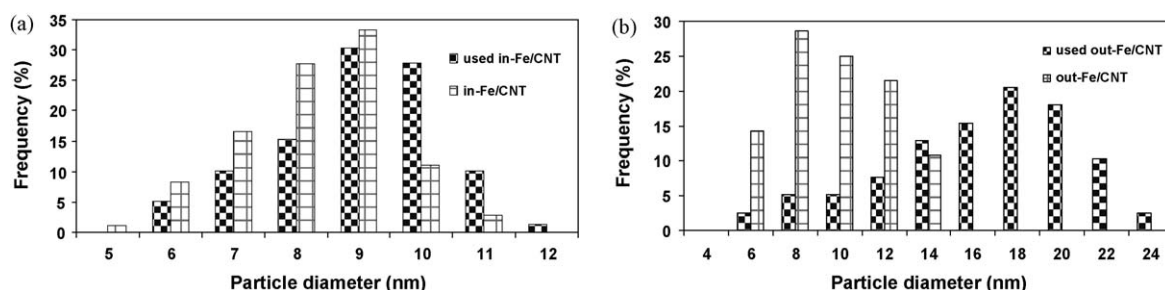


Fig. 6. Particle size distributions for (a) fresh and used in-Fe/CNT, (b) fresh and used out-Fe/CNT.

process. In the case of the coalescence process, the movement of crystallites themselves over the support results in the particle collision and formation of larger particles. In both processes, sintering slows down with time resulting in a stable state [24].

In order to distinguish between these two models, the pattern of particle size distribution after sintering is generally studied. It has been shown that the Ostwald ripening results in a particle size distribution with a tail toward small particle sizes and a steep slope toward larger particle sizes [24,25]. As can be seen in Fig. 6b, there is a Log-normal distribution with a tail toward small particle sizes. Thus, the Ostwald ripening can be considered as the mechanism for metal site agglomeration for iron catalysts supported on the carbon nanotubes. Similar mechanism has been proposed for migration of transition metal over carbon base supports [26].

4. Fischer–Tropsch synthesis

4.1. Catalytic performance vs. time on-stream

The performances of the in-Fe/CNT and out-Fe/CNT catalysts for the Fischer–Tropsch synthesis were evaluated in a fixed-bed reactor. All the reactions were performed under a set of standard conditions (270 °C, 2 MPa, $H_2/CO = 2$). CO hydrogenation was performed on the acid treated CNT support (blank runs with no iron) under the same operating conditions as the metal loaded samples. In the blank runs, the main product formed at a very low conversion (1%) was methane with almost no higher hydrocarbons.

Fig. 7 shows the variations of CO conversion with the time on-stream (TOS). Both catalysts reached their highest activity within 10 h. Afterward, they showed different stability pattern within a time period of 125 h. Fig. 7 shows that in-Fe/CNT is a more stable

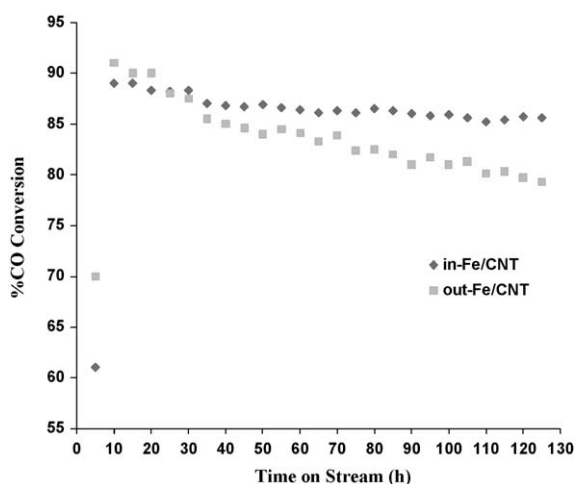


Fig. 7. Change in %CO conversion with time on-stream for in-Fe/CNT and out-Fe/CNT catalysts (Process conditions: 2 Sl/g-cat/h, $P = 2$ MPa, $H_2/CO = 2$, $T = 270$ °C, TOS of 125 h).

catalyst compared to out-Fe/CNT. In the case of in-Fe/CNT, syngas conversion decreased slightly from 89 to 85% in a period of 125 h, whereas, out-Fe/CNT catalyst showed a quick drop in CO conversion from 91 to 79%. As discussed in the previous section, the catalytic site agglomeration during the FT reaction is believed to be the main reason for deactivation of out-Fe/CNT catalyst.

The product selectivities of the catalysts (CH_4 , C_2-C_5 , C_5+ and CO_2) after time on-stream of 125 h are given in Table 4. The data show that CH_4 selectivity of the in-Fe/CNT (25 wt%) catalyst is lower than that of the out-Fe/CNT catalyst (40 wt%). Also C_5+ selectivity of the in-Fe/CNT (36 wt%) is higher than that of the out-Fe/CNT (23 wt%). CO_2 selectivities of the in-Fe/CNT and out-Fe/CNT catalysts are comparable, indicating similar water-gas shift activity for both catalysts. It should be noted that there was a slight shift toward lighter hydrocarbons (3% decrease in C_5+) during the reaction period for the in-Fe/CNT catalyst. In contrary, the out-Fe/CNT catalyst experienced a decrease of 5% in CH_4 selectivity in the course of 125 h. This could be related to drop in CO conversion and the increase in particle sizes.

The results of FT reaction product analysis for both in-Fe/CNT and out-Fe/CNT catalysts showed that the deposition of catalytic metals inside the nanotube pores enhanced product selectivity of the catalyst. Several reasons can be associated with the improvement in product selectivity. As discussed earlier, H_2 -TPR analysis revealed that the reducibility of the in-Fe/CNT catalyst was higher in comparison to the out-Fe/CNT. This phenomenon can result in the formation of more catalytically active carbide species during FTS. In addition, confinement of the reaction intermediates inside the pores can enhance their contact with catalytic sites, favoring the growth of longer chain hydrocarbons. In addition, the inner sides of the CNTs are electron-deficient and can enhance the dissociation of CO resulting in production of higher chain hydrocarbons.

4.2. Effect of temperature on product selectivity

Catalytic activity and product selectivity of in-Fe/CNT and out-Fe/CNT catalysts were analysed at different temperatures ranging from 260 to 285 °C. Fig. 8a and b show %CO conversion and CH_4 selectivity of the in-Fe/CNT and out-Fe/CNT catalysts at a pressure of 2 MPa and space velocity of 4 Sl/g-cat/h and time on-stream of 125 h. As can be seen in Fig. 7a, activity of both catalysts follows similar trends as reaction temperature increases. However, after a

Table 4
Product selectivity (wt%) for in-Fe/CNT and out-Fe/CNT.

Product selectivity	In-Fe/CNT	Out-Fe/CNT
CH_4	25.6	40.5
C_2-C_4	38.2	35.7
C_5+	36.2	23.8
CO_2	38.9	39.5

$T = 270$ °C, $P = 2$ MPa, space velocity = 2 Sl/g-cat/h, $CO/H_2 = 2$. Hydrocarbon selectivity: wt%, CO_2 selectivity (C%), time on-stream of 125 h.

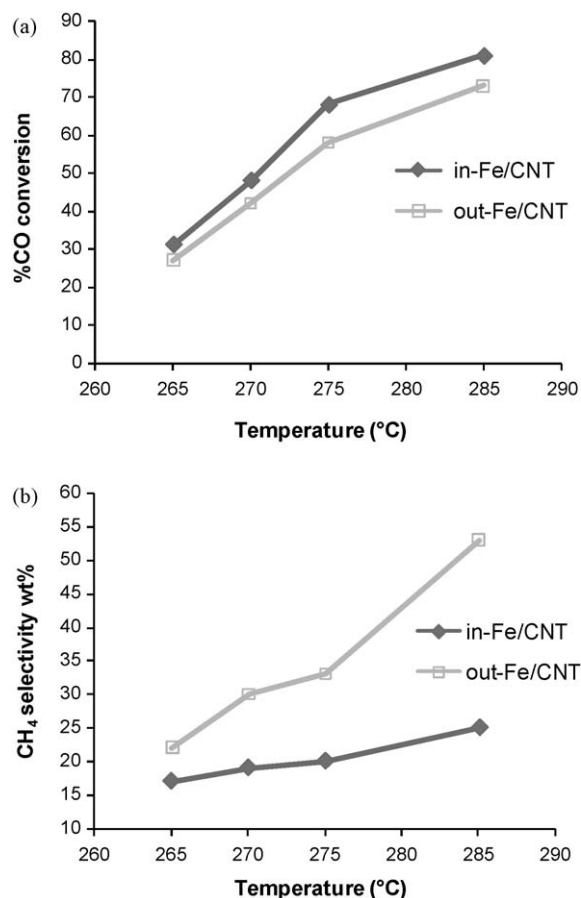


Fig. 8. Change in catalytic performance with reaction temperature for in-Fe/CNT and out-Fe/CNT catalysts (a) %CO conversion, (b) CH₄ selectivity (Process conditions: TOS of 125 h, 4 SL/g-cat/h, $P = 2$ MPa, $H_2/CO = 2$).

reaction period of 125 h, in-Fe/CNT catalyst shows higher CO conversion compared to out-Fe/CNT. The difference between CO conversion for both catalysts is also more pronounced at elevated temperatures. In the case of CH₄ selectivity, which is an undesired by-product of FT reaction; the in-Fe/CNT catalyst exhibits better performance compared to the out-Fe/CNT catalyst. For the out-Fe/CNT catalyst, with an increase in 25 °C in the reaction temperature, the CH₄ selectivity increases from 22 to 53 wt%. However, in the case of the in-Fe/CNT catalyst, there is only an increase in 8% in CH₄ selectivity. Generally, in FTS reactions, an increase in temperature results in a shift toward products with a lower carbon number.

The effect of temperature on the FTS reactions is complex. Kinetics of all reactions including primary reactions, secondary reactions (hydrogenations and isomerizations) and chain termination and the chain growth mechanism can be influenced by

temperature. In addition, physical properties that control the rate of reaction, such as diffusion, depend on the temperature. In the case of the catalyst with active sites inside the nanopores, the longer contact time for reaction intermediates inside the nanotubes channels (i.d. of 8–10 nm and a length of 0.5–2 μ m) can result in a higher selectivity to longer chain hydrocarbons and lower methane selectivity.

5. Conclusions

The results of this research show that the deposition of catalytic sites inside the nanotube pores improves the reduction behavior of the CNT catalyst most likely due to difference in electronic properties of the inner and outer surface of the CNTs and confinement effects. In the case of catalysts with catalytic particles inside the pores, confinement of reaction intermediates inside the channels increases the contact time of the reactants with the active sites, resulting in production of heavier hydrocarbons. Another advantage of deposition of catalytic species inside the pores is the physical encapsulation of catalytic particles, which can reduce the site sintering.

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